Felipe Meneguitti Dias

AI engineering for healthcare

I am a Senior R&D Specialist at Samsung with a PhD from the University of São Paulo. My work spans healthcare AI, signal processing and model validation, from development to deployment.

Selected work

Heart Institute (InCor), Hospital das Clínicas, University of São Paulo Medical School

Gorgona

ECG classification for urgent care units, with processing at InCor and Telegram alerts alongside telemedicine. I led the project and the initial integration work.

ECG classification at InCor

ECG classification integrated with PACS and the emergency department application. I developed models, coordinated clinical annotation and worked on deployment and evaluation.

RPMS

Remote monitoring with watches, phone-based PPG processing and a clinical dashboard. I developed collection protocols, preprocessing and algorithms with the team.

Samsung · 2026–present | InCor · 2020–2026

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Gorgona

ECG classification integrated into urgent care

Objective

Flag ECGs with suspected current of injury and notify the clinical coordination team earlier. Automated analysis runs alongside the telemedicine workflow. Patient assessment and care decisions remain with the medical team.

My contribution

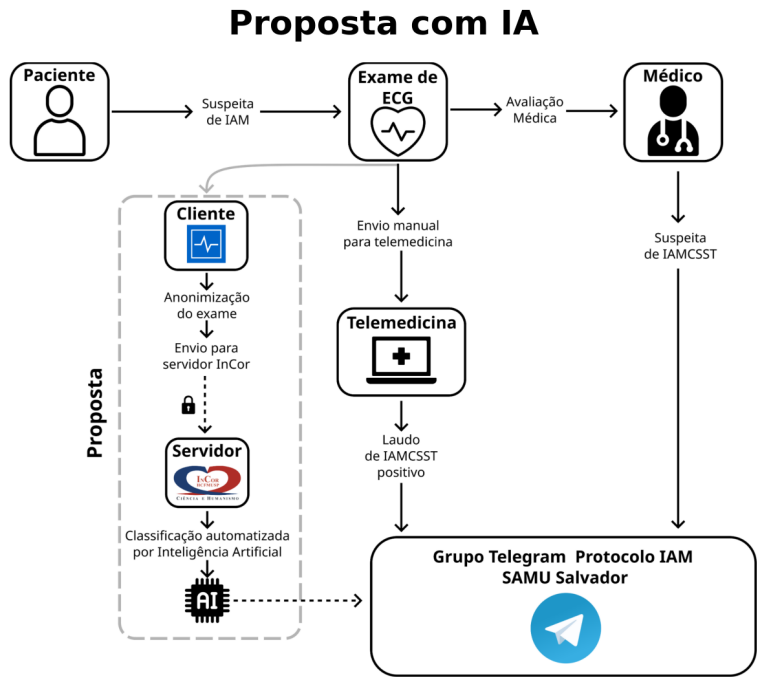
I connected the classification model to the Windows client, InCor processing server, web dashboard and Telegram alerts. The clinical team managed the care protocol and reviewed the ECGs.

Previous workflow

Manual submission to telemedicine and staff communication with the coordination team, based on clinical assessment.

With Gorgona

Windows client, configurable anonymization, InCor server and automated Telegram alert.



Proposed AI workflow. Original project figure, in Portuguese.

Telemedicine and clinical decisions remain part of the workflow. HTTPS and per-installation credentials complement configurable local anonymization.

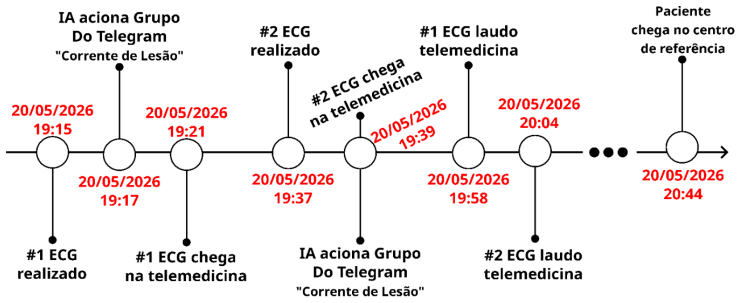
Case study

2 min
ECG to alert in the case

41 min
before the first report

UPA Rodrigo Argolo · 20/05/2026

On May 20, 2026, one encounter at UPA Rodrigo Argolo provided a complete record of the alert workflow. Admission was at 18:52. Both ECGs triggered automated alerts 2 minutes after acquisition. Arrival at the referral center was recorded at 20:44.



Excerpt from the original case-study timeline, slide 23. The crop excludes the personal identification block and starts at the first ECG. Admission: 18:52. Original figure in Portuguese.

The first alert arrived 4 minutes before the ECG reached telemedicine. Alerts preceded the corresponding reports by 41 and 25 minutes. The case demonstrates earlier information delivery to the coordination team. It does not establish a causal reduction in treatment time or improved clinical outcomes.

Classification model

The classifier was developed from the AIMED model that won the 2024 PhysioNet Challenge classification task. The CinC 2024 paper and my doctoral thesis describe the method.

References

Image-Based Electrocardiogram Classification Using Pre-trained ConvNext · Computing in Cardiology, 2024.

Classificação automatizada de imagens de eletrocardiogramas por aprendizado profundo · Universidade de São Paulo · Tese de doutorado, 2025.

Image-based deep learning for emergency electrocardiogram classification · medRxiv · Preprint, 2026.

See the full case study for the location map, Windows client and online platform

ECG classification at InCor

Objective

Support the prioritization of ECG review in the emergency room by bringing automated classification into the clinical team's workspace.

My contribution

I developed and deployed ECG classification models in emergency care. I coordinated the annotation of approximately 20,000 examinations by 19 cardiologists, covering class definitions, adjudication and agreement analysis, as well as post-deployment evaluation and comparison with specialists.

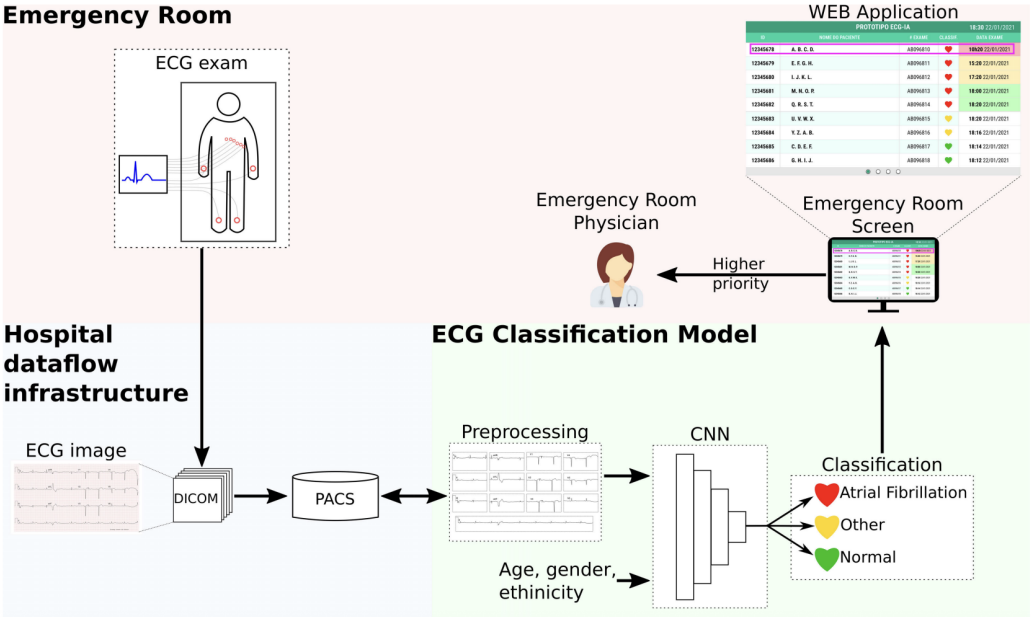


Figure 1 from Dias et al., IEEE Access (2023). Proposed integration of ECG acquisition, DICOM/PACS, classification and the emergency room web application. Image provided by the author.

Results and evaluation

The 2023 architecture uses three classes: atrial fibrillation, other and normal. Subsequent research developed a ConvNeXt ensemble for 12 categories. The 2026 study includes 18,519 ECGs from 14,402 patients and achieved a macro F1 of 0.807 on the held-out test set (95% CI: 0.788–0.825).

Deployment included cardiologist review and comparison with specialists. The research dataset size does not represent routine clinical processing volume.

References

Artificial Intelligence-Driven Screening System for Rapid Image-Based Classification of 12-Lead ECG Exams: A Promising Solution for Emergency Room Prioritization · IEEE Access, 2023.

Image-based deep learning for emergency electrocardiogram classification · medRxiv · Preprint, 2026.

Classificação automatizada de imagens de eletrocardiogramas por aprendizado profundo · Universidade de São Paulo · Tese de doutorado, 2025.

RPMS

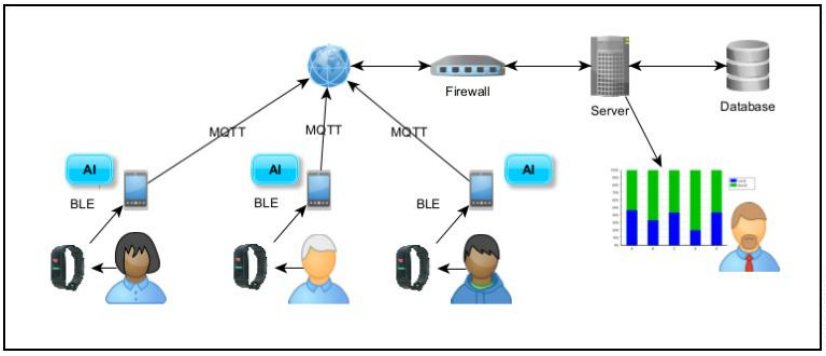
Remote Patient Monitoring System

Objective

Support patient follow-up outside the hospital by connecting wearable recordings to biomarkers available to the clinical team.

My contribution

I developed collection protocols with prototype watches, organized the data and worked on photoplethysmography (PPG) preprocessing and algorithms with the team.



Architecture published in Moreno, Dias et al. (SIoT, 2022), Figure 1. Signals are processed on the phone.

The watch sends PPG to the Android app over Bluetooth Low Energy. The phone processes the signal, estimates blood pressure and heart rate, and sends results over MQTT to a ThingsBoard server. The dashboard displays each patient's measurements over time.

52 volunteers and 15 wristbands in the version published in 2022.

Validation and research

The published protocol keeps participants separate across validation folds. The 2022 implementation estimates blood pressure and heart rate. Signal quality, diabetes and sleep are research areas documented in the papers below.

References

IoT Medical Device Architecture to Estimate Non-invasive Arterial Blood Pressure · Symposium on Internet of Things (SIoT), 2022.

Quality Assessment of Photoplethysmography Signals For Cardiovascular Biomarkers Monitoring Using Wearable Devices · arXiv · Preprint, 2023.

Exploring the limitations of blood pressure estimation using the photoplethysmography signal · Physiological Measurement, 2025.

Machine Learning-Based Diabetes Detection Using Photoplethysmography Signal Features · SBCAS, 2024.

Enhancing Photoplethysmography-Based Sleep Staging Models Through Temporal Context Optimization · Studies in Health Technology and Informatics, 2025.